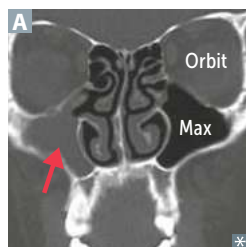


► RESPIRATORY—PATHOLOGY

Rhinosinusitis

Obstruction of sinus drainage into nasal cavity → inflammation and pain over affected area.

Typically affects maxillary sinuses, which drain against gravity due to ostia located superomedially (red arrow points to fluid-filled right maxillary sinus in **A**).

Superior meatus—drains posterior ethmoid; middle meatus—drains frontal, maxillary, and anterior ethmoid; inferior meatus—drains nasolacrimal duct.

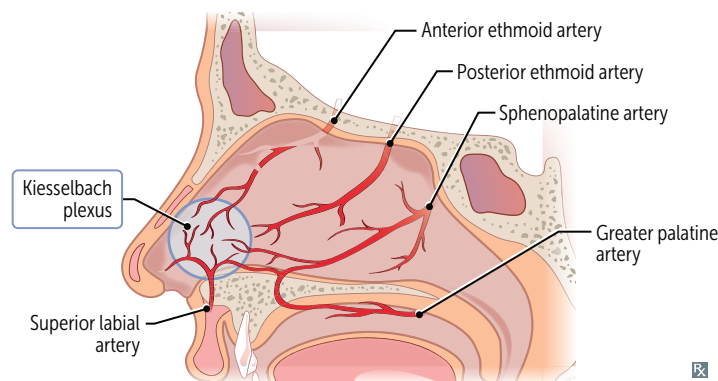
Acute rhinosinusitis is most commonly caused by viruses (eg, rhinovirus); may lead to superimposed bacterial infection, most commonly nontypeable *H influenzae*, *S pneumoniae*, *M catarrhalis*.

Paranasal sinus infections may extend to the orbits, cavernous sinus, and brain, causing complications (eg, orbital cellulitis, cavernous sinus syndrome, meningitis).

Epistaxis

Nose bleed. Most commonly occurs in anterior segment of nostril (**Kiesselbach plexus** at caudal septum). Life-threatening hemorrhages occur in posterior segment (sphenopalatine artery, a branch of maxillary artery). Common causes include foreign body, trauma, allergic rhinitis, and nasal angiofibromas (common in adolescent males).

Kiesselbach drives his **Lexus** with his **LEGS**: superior **L**abial artery, anterior and posterior **E**thmoidal arteries, **G**reater palatine artery, **S**phenopalatine artery.

**Head and neck cancer**

Mostly squamous cell carcinoma. Risk factors include tobacco, alcohol, HPV-16 (oropharyngeal), EBV (nasopharyngeal). Field cancerization: carcinogen damages wide mucosal area → multiple tumors develop independently after exposure.

Nasopharyngeal carcinoma may present with unilateral nasal obstruction, discharge, epistaxis. Eustachian tube obstruction may lead to otitis media +/- effusion, hearing loss.

Laryngeal papillomatosis—also called recurrent respiratory papillomatosis. Benign laryngeal tumor, commonly affecting areas of stratified squamous epithelium such as the true vocal cords, especially in children (possibly from HPV transmitted from mother to baby during labor). Associated with HPV-6 and HPV-11. Symptoms may guide location of pathology (supraglottic → dysphagia, infraglottic/glottic → hoarseness).

Pulmonary emboli

Obstruction of the pulmonary artery or its branches by foreign material (usually thrombus) that originated elsewhere. Affected alveoli are ventilated but not perfused ($\dot{V}/\dot{Q}$ mismatch). May present with sudden-onset dyspnea, pleuritic chest pain, tachypnea, tachycardia, hypoxemia, respiratory alkalosis. Large emboli or saddle embolus (red arrows show filling defects in **A**) may cause sudden death due to clot preventing blood from filling LV and increased RV size further compromising LV filling (obstructive shock). CT pulmonary angiography is imaging test of choice for PE (look for filling defects) **B**. ECG may show sinus tachycardia or, less commonly, S1Q3T3 abnormality. Lines of Zahn **C** are interdigitating areas of pink (platelets, fibrin) and red (RBCs) found only in thrombi formed before death; help distinguish pre- and postmortem thrombi.

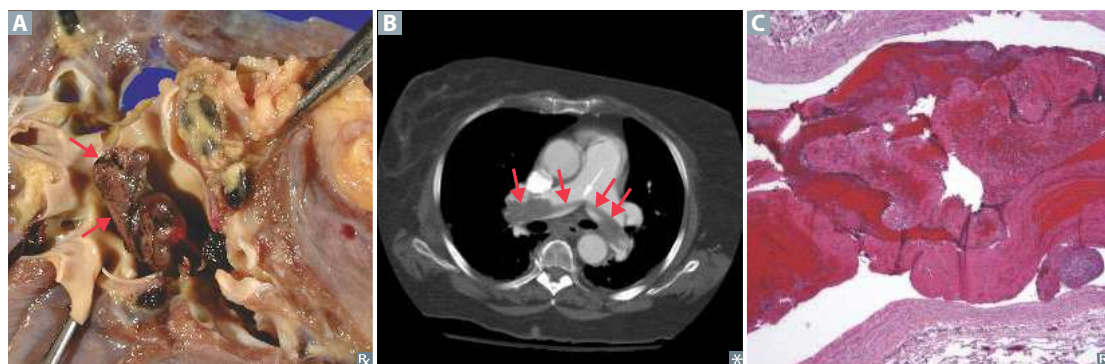
Treatment: anticoagulation (eg, heparin, direct thrombin/factor Xa inhibitors), IVC filter (if anticoagulation is contraindicated).

Types: **F**at, **A**ir, **T**hrombus, **B**acteria, **A**mniotic fluid, **T**umor. An embolus moves like a **FAT BAT**.

Fat emboli—associated with long bone fractures and liposuction; classic triad of hypoxemia, neurologic abnormalities, petechial rash.

Air emboli—nitrogen bubbles precipitate in ascending divers (caisson disease/decompression sickness); treat with hyperbaric O_2 ; or, can be iatrogenic 2° to invasive procedures (eg, central line placement).

Amniotic fluid emboli—typically occurs during labor or postpartum, but can be due to uterine trauma. Can lead to DIC. Rare, but high mortality.

**Mediastinal pathology**

Normal mediastinum contains heart, thymus, lymph nodes, esophagus, and aorta.

Mediastinal masses

Some pathologies (eg, lymphoma, lung cancer, abscess) can occur in any compartment, but there are common associations:

- Anterior—**4 T**'s: **t**hyroid (substernal goiter), **t**hymic neoplasm, **t**eratoma, "**t**errible" lymphoma.
- Middle—metastases, hiatal hernia, bronchogenic cysts.
- Posterior—esophageal cancer (may present as mass in, or spread to, middle mediastinum), neurogenic tumor (eg, neurofibroma), multiple myeloma.

Mediastinitis

Inflammation of mediastinal tissues. Commonly due to postoperative complications of cardiothoracic procedures (≤ 14 days), esophageal perforation (common with repetitive vomiting), or contiguous spread of odontogenic/retropharyngeal infection.

Chronic mediastinitis—also called fibrosing mediastinitis; due to $\uparrow$ proliferation of connective tissue in mediastinum. *Histoplasma capsulatum* is common cause.

Clinical features: fever, tachycardia, leukocytosis, chest pain, and sternal wound drainage.

Pneumomediastinum

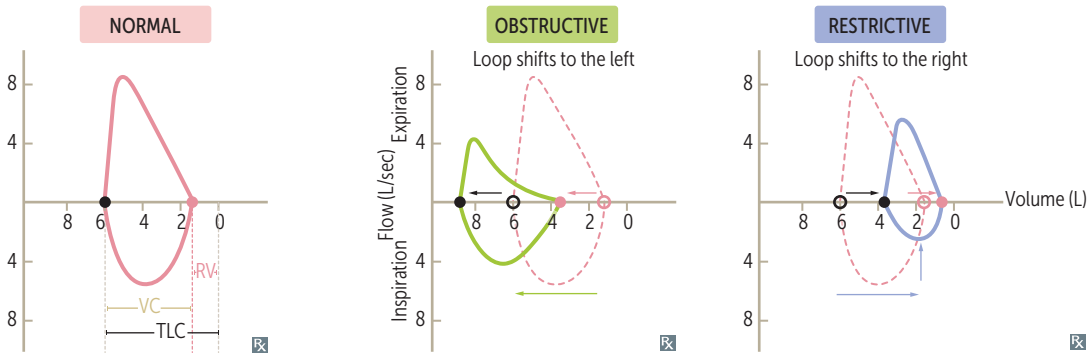
Presence of gas (usually air) in the mediastinum. Can either be spontaneous (due to rupture of pulmonary bleb) or 2° (eg, trauma, iatrogenic, Boerhaave syndrome).

Ruptured alveoli allow tracking of air into the mediastinum via peribronchial and perivascular sheaths.

Clinical features: chest pain, dyspnea, voice change, subcutaneous emphysema, $\oplus$ Hamman sign (crepitus on cardiac auscultation).

Flow-volume loops

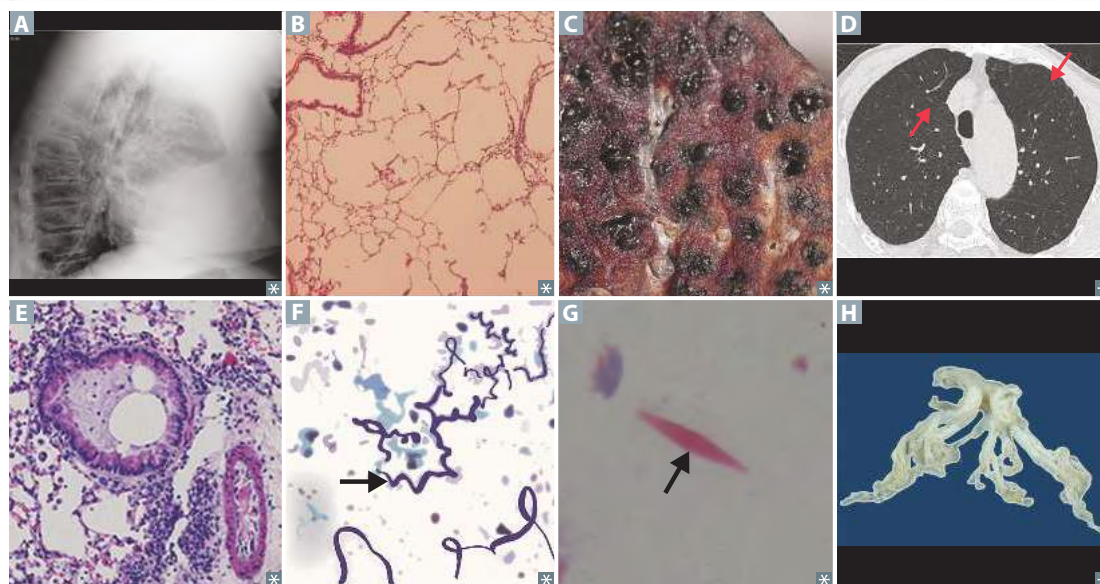
FLOW-VOLUME PARAMETER	Normal	Obstructive lung disease	Restrictive lung disease
RV		↑	↓
FRC		↑	↓
TLC		↑	↓
FEV ₁	>80% predicted	↓↓	↓
FVC	>80% predicted	Normal or ↓	↓
FEV ₁ /FVC	>70%	↓ FEV ₁ decreased more than FVC	Normal or ↑ FEV ₁ decreased proportionately to FVC



Obstructive lung diseases	Obstruction of air flow (↓↓ FEV ₁ , ↓ FVC ↓ FEV ₁ /FVC ratio) → air trapping in lungs (↑ RV, →↑ FRC and ↑ TLC) due to premature airway closure at high lung volumes. Includes COPD (chronic bronchitis and emphysema), asthma, and bronchiectasis.
Chronic obstructive pulmonary disease	Often due to tobacco use (most important risk factor), pollutants, or allergens. Includes chronic bronchitis and emphysema, which often co-exist. Exacerbation: acute worsening of symptoms, often associated with viral or bacterial upper respiratory tract infection.
Chronic bronchitis	
DIAGNOSIS	Clinical diagnosis. Criteria: productive cough for ≥ 3 months in ≥ 2 consecutive years. May also have dyspnea, wheezes, crackles (due to mucus), cyanosis (hypoxemia due to shunting), 2° polycythemia. Leads to metaplasia of pseudostratified ciliated columnar epithelium into stratified squamous epithelium.
MECHANISMS	Hypertrophy and hyperplasia of mucus-secreting glands in bronchi.
NOTES	↑ Reid index (thickness of mucosal gland layer to thickness of wall between epithelium and cartilage) > 50%.
Emphysema	
DIAGNOSIS	Radiologic or biopsy diagnosis. CXR: barrel chest, ↑ AP diameter (best seen in lateral A), flattened diaphragm, ↑ lung field lucency.
MECHANISMS	Alveolar wall destruction B → ↑ compliance of lung, ↓ recoil, and damage to alveolar capillary membrane → ↓ DLCO; results in ↑ air space. Centriacinar—spares distal alveoli, frequently in upper lobes. Associated with tobacco smoking C D . Panacinar—affects respiratory bronchioles and alveoli, frequently in lower lobes. Associated with α ₁ -antitrypsin deficiency.

Obstructive lung diseases (continued)

NOTES	Mediated by oxidative stress, chronic inflammation (CD8+ T cells, neutrophils, and macrophages), and imbalance of proteases and antiproteases ($\uparrow$ elastase activity $\rightarrow$ $\uparrow$ loss of elastic fibers $\rightarrow$ alveolar destruction). Defect/deficiency/absence of α_1 -antitrypsin (antiprotease that inhibits neutrophil elastase) leads to unopposed elastase activity.
Asthma	Intermittent obstructive lung disease often triggered by allergens, viral URIs, stress. Associated with atopy. NSAID- or aspirin-exacerbated respiratory disease—asthma, nasal polyps, and COX-inhibitor sensitivity (leukotriene overproduction $\rightarrow$ airway constriction) (Samter's triad).
DIAGNOSIS	Clinical diagnosis. Intermittent episodes of dyspnea, coughing, wheezing, tachypnea. Diagnosis supported by spirometry (obstructive pattern with bronchodilator response, but may be normal when not in exacerbation) $\pm$ methacholine challenge.
MECHANISMS	Type I hypersensitivity reaction $\rightarrow$ smooth muscle hypertrophy and hyperplasia. Hyperresponsive bronchi $\rightarrow$ reversible bronchoconstriction. Mucus plugging E .
OTHER	Curschmann spirals F —shed epithelium forms whorled mucus plugs. Charcot-Leyden crystals G —eosinophilic, hexagonal, double-pointed crystals formed from breakdown of eosinophils in sputum.
Bronchiectasis	Obstructive lung disease. Most commonly associated with cystic fibrosis.
DIAGNOSIS	Characterized by chronic cough and daily purulent sputum production. Often have recurrent pulmonary infections. Confirmed by imaging demonstrating airway dilation and bronchial thickening. Supported by obstructive PFT pattern.
PATHOPHYSIOLOGY	Initial insult of pulmonary infection combined with obstruction or impaired clearance $\rightarrow$ dysregulated host response $\rightarrow$ bronchial inflammation $\rightarrow$ permanently dilated airways.
NOTES	Many etiologies, including airway obstruction (eg, foreign body aspiration, mass), poor ciliary motility (eg, tobacco smoking, Kartagener syndrome), cystic fibrosis (H shows a coughed up inspissated mucus plug), allergic bronchopulmonary aspergillosis, pulmonary infections (eg, <i>Mycobacterium avium</i>).



Restrictive lung diseases

May lead to ↓ lung volumes (↓ FVC and TLC). PFTs: normal or ↑ FEV₁/FVC ratio. Patient presents with short, shallow breaths, crackles (velcro-type).

Types:

- Altered respiratory mechanics (extrapulmonary, normal D_{LCO}, normal A-a gradient):
 - Respiratory muscle weakness—polio, myasthenia gravis, Guillain-Barré syndrome, ALS
 - Chest wall abnormalities—scoliosis, severe obesity
- Diffuse parenchymal lung diseases, also called interstitial lung diseases (pulmonary, ↓ D_{LCO}, ↑ A-a gradient):
 - Pneumoconioses (eg, coal workers' pneumoconiosis, silicosis, asbestosis)
 - Sarcoidosis: bilateral hilar lymphadenopathy, noncaseating granulomas; ↑ ACE and Ca²⁺
 - Idiopathic pulmonary fibrosis
 - Granulomatosis with polyangiitis
 - Pulmonary Langerhans cell histiocytosis (eosinophilic granuloma)
 - Hypersensitivity pneumonitis
 - Drug toxicity (eg, bleomycin, busulfan, amiodarone, methotrexate)
 - Acute respiratory distress syndrome
 - **Radiation-induced lung injury**—associated with proinflammatory cytokine release (eg, TNF-α, IL-1, IL-6). May be asymptomatic but most common symptoms are dry cough and dyspnea +/- low-grade fever. Acute radiation pneumonitis develops within 3–12 weeks (exudative phase); radiation fibrosis may develop after 6–12 months.

Idiopathic pulmonary fibrosis

Progressive fibrotic lung disease of unknown etiology. May involve multiple cycles of lung injury, inflammation, and fibrosis. Associated with tobacco smoking, environmental pollutants, genetic defects.

Findings: progressive dyspnea, fatigue, nonproductive cough, crackles, clubbing. Imaging shows peripheral reticular opacities with traction bronchiectasis +/- “honeycomb” appearance of lung (advanced disease). Histologic pattern: usual interstitial pneumonia. ↓ type 1 pneumocytes, ↑ type 2 pneumocytes, ↑ fibroblasts.

Complications: pulmonary hypertension, right heart failure, arrhythmias, coronary artery disease, respiratory failure, lung cancer.

Hypersensitivity pneumonitis

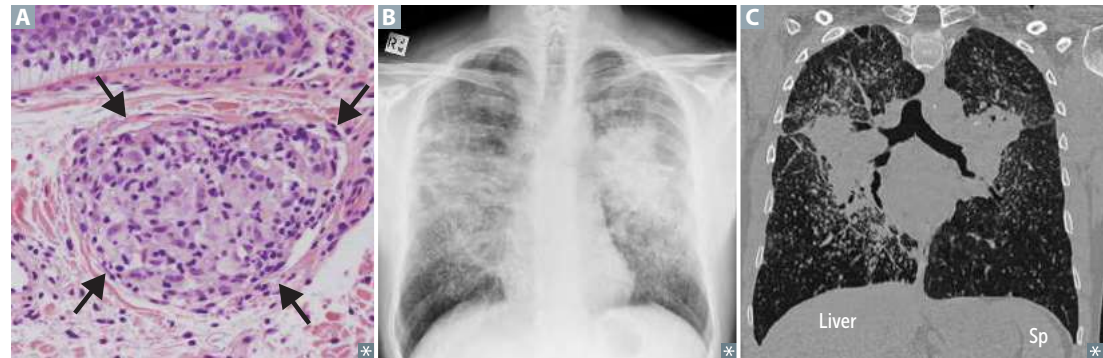
Mixed type III/IV hypersensitivity reaction to environmental antigens such as thermophilic *Actinomyces* and *Aspergillus*. Often seen in farmers and bird-fanciers. Acutely, causes dyspnea, cough, chest tightness, fever, headache. Often self-limiting if stimulus is removed. Chronically, leads to irreversible fibrosis with noncaseating granuloma, alveolar septal thickening, traction bronchiectasis.

Sarcoidosis

Characterized by immune-mediated, widespread noncaseating granulomas **A**, elevated serum ACE levels, and elevated CD4/CD8 ratio in bronchoalveolar lavage fluid. More common in Black females. Often asymptomatic except for enlarged lymph nodes. CXR shows bilateral adenopathy and coarse reticular opacities **B**, including ground glass opacities; CT of the chest better demonstrates the extensive hilar and mediastinal adenopathy **C**.

Associated with Bell palsy, parotid enlargement, granulomas (noncaseating epithelioid, containing microscopic Schaumann and Asteroid bodies), Rheumatoid arthritis–like arthropathy, ↑ Calcium, Ocular uveitis, Interstitial fibrosis, vitamin D activation (due to ↑ 1α -hydroxylase in macrophages), Skin changes (eg, lupus pernio, erythema nodosum) (**SARCOIDS**).

Treatment: glucocorticoids (if symptomatic).

**Mesothelioma**

Malignancy of the pleura associated with asbestosis. May result in hemorrhagic pleural effusion (exudative), pleural thickening.

Histology may show psammoma bodies.

EM may show polygonal tumor cells with microvilli, desmosomes, tonofilaments.

Calretinin and cytokeratin 5/6 ⊕ in almost all mesotheliomas, ⊖ in most carcinomas.

Tobacco smoking is not a risk factor.

Pneumoconioses

Asbestos is from the **roof** (was common in insulation), but affects the **base** (lower lobes).

Silica, **coal**, and **berries** are from the **base** (earth), but affect the **roof** (upper lobes).

Asbestos-related disease

Asbestos causes asbestosis (pulmonary fibrosis), pleural disease, malignancies. Associated with shipbuilding, roofing, plumbing. “Ivory white,” calcified, supradiaphragmatic and pleural **A** plaques are pathognomonic.

Risk of bronchogenic carcinoma > risk of mesothelioma. ↑ risk of Caplan syndrome (rheumatoid arthritis and pneumoconioses with intrapulmonary nodules).

Affects lower lobes.

Asbestos (ferruginous) bodies are golden-brown fusiform rods resembling dumbbells, found in alveolar sputum sample, visualized using Prussian blue stain **B**, often obtained by bronchoalveolar lavage.

↑ risk of pleural effusions.

Berylliosis

Associated with exposure to beryllium in aerospace and manufacturing industries. Granulomatous (noncaseating) **C** on histology and therefore occasionally responsive to glucocorticoids. ↑ risk of cancer and cor pulmonale.

Affects upper lobes.

Coal workers' pneumoconiosis

Prolonged coal dust exposure → macrophages laden with **carbon** → inflammation and fibrosis.

Also called black lung disease. ↑ risk of **Caplan** syndrome.

Affects upper lobes.

Small, rounded nodular opacities seen on imaging.

Anthracosis—asymptomatic condition found in many urban dwellers exposed to sooty air.

Silicosis

Associated with **sand**blasting, **foundries**, **mines**. Macrophages respond to silica and release fibrogenic factors, leading to fibrosis. It is thought that silica may disrupt phagolysosomes and impair macrophages, increasing susceptibility to TB. ↑ risk of lung cancer, cor pulmonale, and Caplan syndrome.

Affects upper lobes.

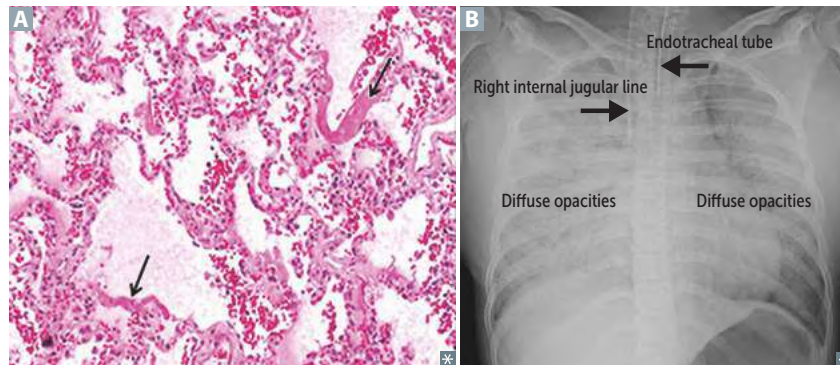
“**Eggshell**” calcification of hilar lymph nodes on CXR.

The **silly egg sandwich** I **found** is **mine**!



Acute respiratory distress syndrome

PATHOPHYSIOLOGY	Alveolar insult → release of pro-inflammatory cytokines → neutrophil recruitment, activation, and release of toxic mediators (eg, reactive oxygen species, proteases, etc) → capillary endothelial damage and ↑ vessel permeability → leakage of protein-rich fluid into alveoli → formation of intra-alveolar hyaline membranes (arrows in A) and noncardiogenic pulmonary edema (normal PCWP) → ↓ compliance and $\dot{V}/\dot{Q}$ mismatch → hypoxic vasoconstriction → ↑ pulmonary vascular resistance. Loss of surfactant also contributes to alveolar collapse (eg, preterm infants, drowning).
CAUSES	Sepsis (most common), aspiration pneumonia, burns, trauma, pancreatitis, drowning injuries.
DIAGNOSIS	Diagnosis of exclusion with the following criteria (ARDS): ▪ A Abnormal chest X-ray (bilateral lung opacities) B ▪ R Respiratory failure within 1 week of alveolar insult ▪ D Decreased $\text{PaO}_2/\text{FiO}_2$ (ratio < 300, hypoxemia due to ↑ intrapulmonary shunting and diffusion abnormalities) ▪ S Symptoms of respiratory failure are not due to HF/fluid overload
CONSEQUENCES	Impaired gas exchange, ↓ lung compliance; pulmonary hypertension.
MANAGEMENT	Treat the underlying cause. Mechanical ventilation: ↓ tidal volume, ↑ PEEP (keeps alveoli open during expiration).

**Sleep apnea**

Repeated cessation of breathing > 10 seconds during sleep → disrupted sleep → daytime somnolence. Diagnosis confirmed by sleep study (polysomnography).
 Nocturnal hypoxia → systemic and pulmonary hypertension, arrhythmias (atrial fibrillation/flutter), sudden death.
 Hypoxia → ↑ EPO release → ↑ erythropoiesis.

Obstructive sleep apnea	Respiratory effort against airway obstruction. PaO_2 is usually normal during the day. Associated with obesity, loud snoring, daytime sleepiness. Usually caused by excess parapharyngeal/oropharyngeal tissue in adults, adenotonsillar hypertrophy in children. Treatment: weight loss, CPAP, dental devices, hypoglossal nerve stimulation, upper airway surgery.
Central sleep apnea	Impaired respiratory effort due to C NS injury/toxicity, C ongestive HF, opioids. May be associated with C heyne-Stokes respirations (oscillations between apnea and hyperpnea). Treatment: positive airway pressure.
Obesity hypoventilation syndrome	Also called Pickwickian syndrome. Obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) → hypoventilation → ↑ PaCO_2 during waking hours (retention); ↓ PaO_2 and ↑ PaCO_2 during sleep. Treatment: weight loss, positive airway pressure.

Pulmonary hypertension

Elevated mean pulmonary artery pressure (> 20 mm Hg) at rest. Results in arteriosclerosis, medial hypertrophy, intimal fibrosis of pulmonary arteries, plexiform lesions. $\uparrow$ pulmonary vascular resistance $\rightarrow \uparrow$ RV pressure $\rightarrow$ RVH (parasternal heave on examination), RV failure.

ETIOLOGIES

Pulmonary arterial hypertension (group 1)

Often idiopathic. Females $>$ males. Heritable PAH can be due to an inactivating mutation in *BMPT2* gene (normally inhibits vascular smooth muscle proliferation); poor prognosis. Pulmonary vasculature endothelial dysfunction results in $\uparrow$ vasoconstrictors (eg, endothelin) and $\downarrow$ vasodilators (eg, NO and prostacyclins).

Other causes include drugs (eg, amphetamines, cocaine), connective tissue disease, HIV infection, portal hypertension, congenital heart disease, schistosomiasis.

Left heart disease (group 2)

Causes include systolic/diastolic dysfunction and valvular disease.

Lung diseases or hypoxia (group 3)

Destruction of lung parenchyma (eg, COPD), lung inflammation/fibrosis (eg, interstitial lung diseases), hypoxemic vasoconstriction (eg, obstructive sleep apnea, living in high altitude).

Chronic thromboembolic (group 4)

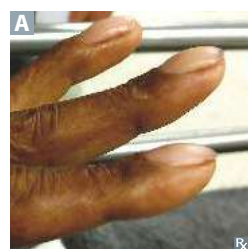
Recurrent microthrombi $\rightarrow \downarrow$ cross-sectional area of pulmonary vascular bed.

Multifactorial (group 5)

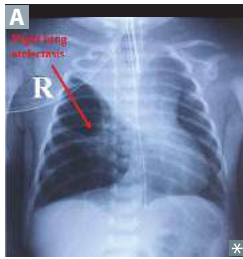
Causes include hematologic, systemic, and metabolic disorders, along with compression of the pulmonary vasculature by a tumor.

Physical findings in select lung diseases

ABNORMALITY	BREATH SOUNDS	PERCUSSION	FREMITUS	TRACHEAL DEVIATION
Pleural effusion	$\downarrow$	Dull	$\downarrow$	None if small Away from side of lesion if large
Atelectasis	$\downarrow$	Dull	$\downarrow$	Toward side of lesion
Simple pneumothorax	$\downarrow$	Hyperresonant	$\downarrow$	None
Tension pneumothorax	$\downarrow$	Hyperresonant	$\downarrow$	Away from side of lesion
Consolidation (lobar pneumonia, pulmonary edema)	Bronchial breath sounds; late inspiratory crackles, egophony, whispered pectoriloquy	Dull	$\uparrow$	None

Digital clubbing

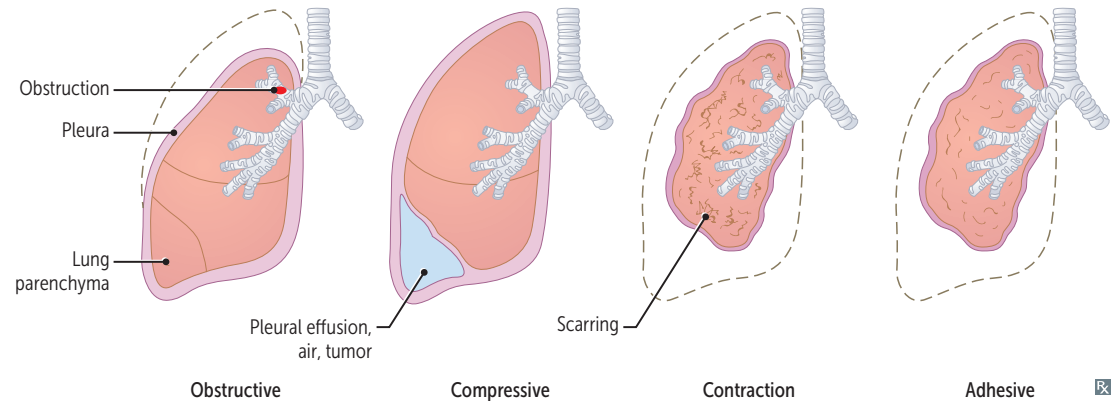
Increased angle between nail bed and nail plate ($> 180^\circ$) **A**. Pathophysiology not well understood; in patients with intrapulmonary shunt, platelets and megakaryocytes become lodged in digital vasculature $\rightarrow$ local release of PDGF and VEGF. Can be hereditary or acquired. Causes include respiratory diseases (eg, idiopathic pulmonary fibrosis, cystic fibrosis, bronchiectasis, lung cancer), cardiovascular diseases (eg, cyanotic congenital heart disease), infections (eg, lung abscess, TB), and others (eg, IBD). Not typically associated with COPD or asthma.

Atelectasis

Alveolar collapse (right upper lobe collapse against mediastinum in **A**). Multiple causes:

- Obstructive—airway obstruction prevents new air from reaching distal airways, old air is resorbed (eg, foreign body, mucous plug, tumor)
- Compressive—external compression on lung decreases lung volumes (eg, space-occupying lesion, pleural effusion)
- Contraction (cicatrization)—scarring of lung parenchyma that distorts alveoli (eg, sarcoidosis)
- Adhesive—due to lack of surfactant (eg, NRDS in premature infants)

Decreased via incentive spirometry or ↑ PEEP during mechanical ventilation.

**Pleural effusions**

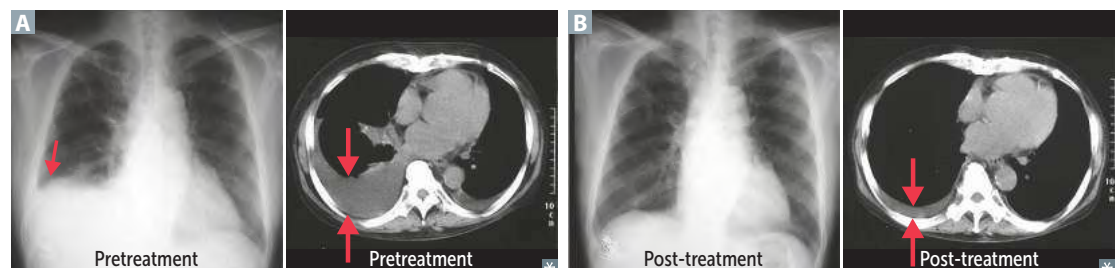
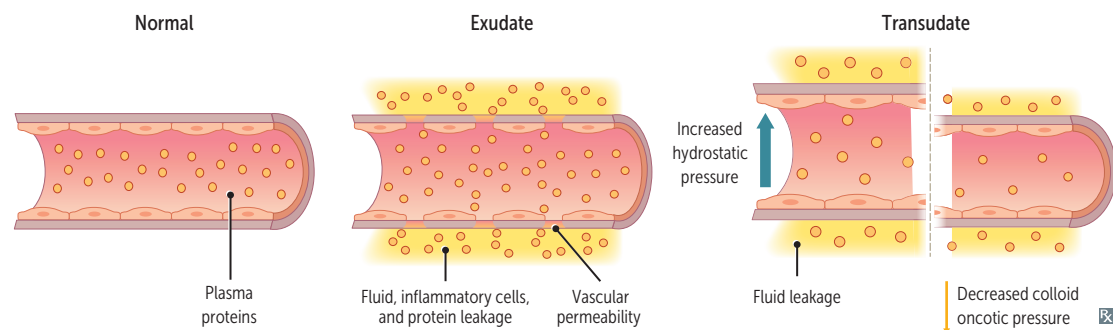
Excess accumulation of fluid **A** between pleural layers → restricted lung expansion during inspiration. Can be treated with thoracentesis to remove/reduce fluid **B**. Based on the Light's criteria, fluid is consistent with an exudate if pleural fluid protein/serum protein > 0.5, pleural fluid LDH/serum LDH > 0.6, or pleural fluid LDH > 2/3 upper limit of normal serum LDH.

Exudate

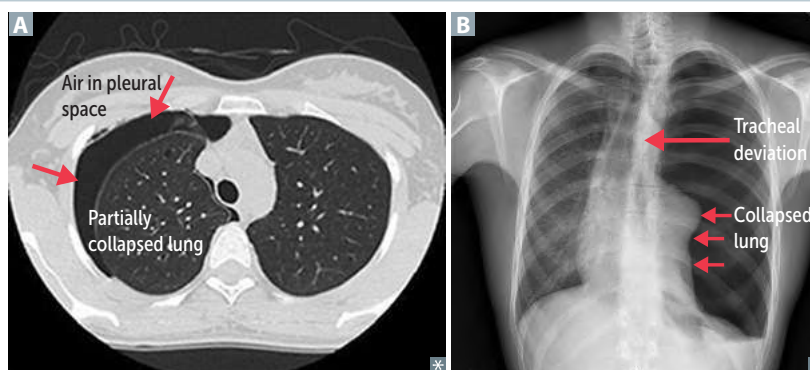
Cloudy fluid (cellular). Due to infection (eg, pneumonia, tuberculosis), malignancy, connective tissue disease, lymphatic (chylothorax), trauma. Often requires drainage due to ↑ risk of infection.

Transudate

Clear fluid (hypocellular). Due to ↑ hydrostatic pressure (eg, HF, Na⁺ retention) and/or ↓ oncotic pressure (eg, nephrotic syndrome, cirrhosis).



Pneumothorax	Accumulation of air in pleural space A . Dyspnea, uneven chest expansion. Chest pain, ↓ tactile fremitus, hyperresonance, and diminished breath sounds, all on the affected side.
Primary spontaneous pneumothorax	Due to rupture of apical subpleural bleb or cysts. Occurs most frequently in tall, thin, young males. Associated with tobacco smoking and vaping.
Secondary spontaneous pneumothorax	Due to diseased lung (eg, bullae in emphysema, Marfan syndrome, infections), mechanical ventilation with use of high pressures → barotrauma.
Traumatic pneumothorax	Caused by blunt (eg, rib fracture), penetrating (eg, gunshot), or iatrogenic (eg, central line placement, lung biopsy, barotrauma due to mechanical ventilation) trauma.
Tension pneumothorax	Can be from any of the above. Air enters pleural space but cannot exit. Increasing trapped air → tension pneumothorax. Trachea deviates away from affected lung B . May lead to increased intrathoracic pressure → mediastinal displacement → kinking of IVC → ↓ venous return → ↓ cardiac output, obstructive shock (hypotension, tachycardia), jugular venous distention. Needs immediate needle decompression and chest tube placement.



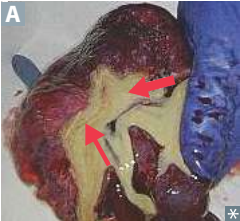
Pneumonia

TYPE	TYPICAL ORGANISMS	CHARACTERISTICS
Lobar pneumonia 	<i>S pneumoniae</i> (most common), <i>Legionella</i> , <i>Klebsiella</i> .	Intra-alveolar exudate → consolidation A ; may involve entire lobe or the whole lung.
Bronchopneumonia	<i>S pneumoniae</i> , <i>S aureus</i> , <i>H influenzae</i> , <i>Klebsiella</i> .	Acute inflammatory infiltrates from bronchioles into adjacent alveoli; patchy distribution involving ≥ 1 lobe.
Interstitial (atypical) pneumonia 	<i>Mycoplasma</i> , <i>Chlamydophila pneumoniae</i> , <i>Chlamydophila psittaci</i> , <i>Legionella</i> , <i>Coxiella burnetii</i> , viruses (RSV, CMV, influenza, adenovirus).	Diffuse patchy inflammation localized to interstitial areas at alveolar walls; CXR shows bilateral multifocal opacities B . Generally follows a more indolent course (“walking” pneumonia).
Cryptogenic organizing pneumonia	Etiology unknown. ⊖ sputum and blood cultures, often responds to glucocorticoids but not to antibiotics.	Formerly called bronchiolitis obliterans organizing pneumonia (BOOP). Noninfectious pneumonia characterized by inflammation of bronchioles and surrounding structure.
Aspiration pneumonia	Aspiration of oropharyngeal or gastric contents → pulmonary infection. Risk factors: altered mental status (↓ cough reflex or glottic closure), dysphagia, neurologic disorders (eg, stroke), invasive tubes (eg, nasogastric tube).	Presents days after aspiration event in dependent lung segment. More common in RLL if sitting up and RUL if lying down (recumbent) due to bronchial anatomy. Can progress to abscess. Aspiration (chemical) pneumonitis —presents hours after aspiration event. Due to gastric acid-mediated inflammation. Presents with infiltrates in lower lobe(s) and resolves with supportive treatment.

Natural history of lobar pneumonia

	Congestion	Red hepatization	Gray hepatization	Resolution
DAYS	1–2	3–4	5–7	8+
FINDINGS	Red-purple, partial consolidation of parenchyma Exudate with mostly bacteria	Red-brown consolidation Exudate with fibrin, bacteria, RBCs, WBCs Reversible	Uniformly gray Exudate full of WBCs, lysed RBCs, and fibrin	Enzymatic digestion of exudate by macrophages

Lung abscess



Localized collection of pus within parenchyma. Caused by aspiration of oropharyngeal contents (especially in patients predisposed to loss of consciousness [eg, alcohol overuse, epilepsy]) or bronchial obstruction (eg, cancer). Air-fluid levels often seen on CXR; presence suggests cavitation. Due to anaerobes (eg, *Bacteroides*, *Fusobacterium*, *Peptostreptococcus*) or *S aureus*. Treatment: antibiotics, drainage, or surgery.

Lung abscess **A** 2° to aspiration is most often found in right lung. Location depends on patient’s position during aspiration: RLL if upright, RUL or RML if recumbent.

Lung cancer

Leading cause of cancer death.

Presentation: cough, hemoptysis, bronchial obstruction, wheezing, pneumonic “coin” lesion on CXR or noncalcified nodule on CT.

Sites of metastases from lung cancer: **liver** (jaundice, hepatomegaly), **adrenals**, **bone** (pathologic fracture), **brain**; “Lung ‘mets’ **Love** affective **bone**heads and **brain**iacs.”

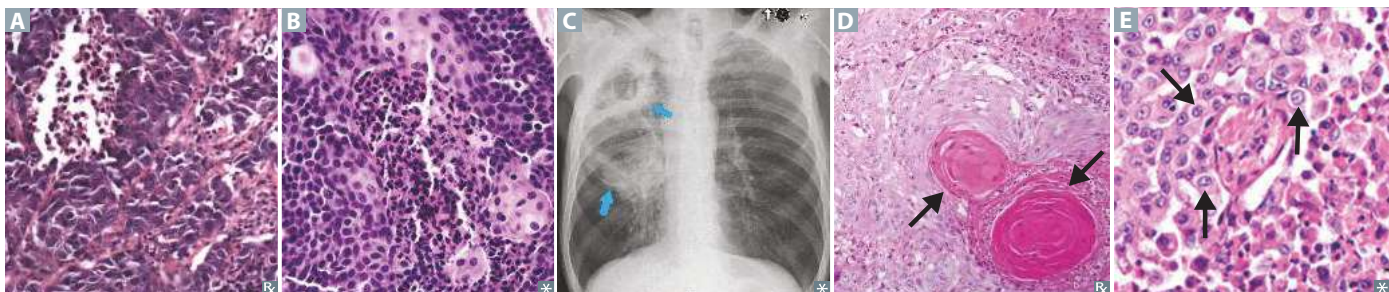
In the lung, metastases (usually multiple lesions) are more common than 1° neoplasms. Most often from breast, colon, prostate, and bladder cancer.

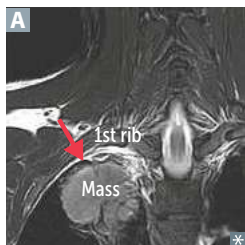
SPHERE of complications: **S**uperior vena cava/thoracic outlet syndromes, **P**ancoast tumor, **H**orner syndrome, **E**ndocrine (paraneoplastic), **R**ecurrent laryngeal nerve compression (hoarseness), **E**ffusions (pleural or pericardial).

Risk factors include tobacco smoking, secondhand smoke, radiation, environmental exposures (eg, radon, asbestos), pulmonary fibrosis, family history.

Squamous and **s**mall cell carcinomas are **s**entral (central) and often caused by tobacco **s**moking. Hamartomas are found incidentally on imaging, appearing as well-circumscribed mass.

TYPE	LOCATION	CHARACTERISTICS	HISTOLOGY
Small cell			
Small cell (oat cell) carcinoma	Central	Undifferentiated → very aggressive. May cause neurologic paraneoplastic syndromes (eg, Lambert-Eaton myasthenic syndrome, paraneoplastic myelitis, encephalitis, subacute cerebellar degeneration) and endocrine paraneoplastic syndromes (Cushing syndrome, SIADH). Amplification of <i>myc</i> oncogenes common. Managed with chemotherapy +/- radiation.	Neoplasm of neuroendocrine Kulchitsky cells → small dark blue cells A . Chromogranin A ⊕, neuron-specific enolase ⊕, synaptophysin ⊕.
Non-small cell			
Adenocarcinoma	Peripheral	Most common 1° lung cancer. Most common subtype in people who do not smoke. More common in females than males. Activating mutations include <i>KRAS</i> , <i>EGFR</i> , and <i>ALK</i> . Associated with hypertrophic osteoarthropathy (clubbing). Bronchioloalveolar subtype (adenocarcinoma in situ): CXR often shows hazy infiltrates similar to pneumonia; better prognosis.	Glandular pattern, often stains mucin ⊕ B . Bronchioloalveolar subtype: grows along alveolar septa → apparent “thickening” of alveolar walls. Tall, columnar cells containing mucus.
Squamous cell carcinoma	C entral	Hilar mass C arising from bronchus; c avitation; c igarettes; hyper c alcemia (produces PTHrP).	Keratin pearls D and intercellular bridges (desmosomes).
Large cell carcinoma	Peripheral	Highly anaplastic undifferentiated tumor. Strong association with tobacco smoking. May produce hCG → gynecomastia (enlarge d breasts). Less responsive to chemotherapy; removed surgically. Poor prognosis.	Pleomorphic giant cells E .
Bronchial carcinoid tumor	Central or peripheral	Excellent prognosis; metastasis rare. Symptoms due to mass effect (wheezing) or carcinoid syndrome (flushing, diarrhea).	Nests of neuroendocrine cells; chromogranin A ⊕.

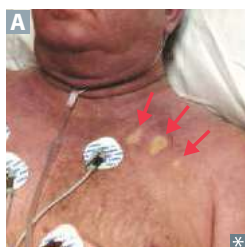


Pancoast tumor

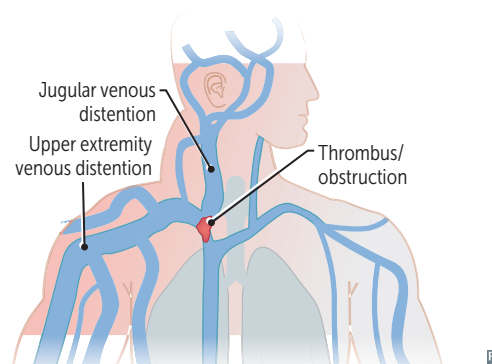
Also called superior sulcus tumor. Carcinoma (most commonly NSCLC) that occurs in the apex of lung **A** may cause Pancoast syndrome by invading/compressing local structures.

Compression of locoregional structures may cause array of findings:

- Recurrent laryngeal nerve → hoarseness
- Stellate ganglion → Horner syndrome (ipsilateral ptosis, miosis, anhidrosis)
- Superior vena cava → SVC syndrome
- Brachiocephalic vein → brachiocephalic syndrome (unilateral symptoms)
- Brachial plexus → shoulder pain, sensorimotor deficits (eg, atrophy of intrinsic muscles of the hand)
- Phrenic nerve → hemidiaphragm paralysis (hemidiaphragm elevation on CXR)

Superior vena cava syndrome

Obstruction of the SVC (eg, thrombus, tumor) impairs blood drainage from the head (“facial plethora”; note blanching after fingertip pressure in **A**), neck (jugular venous distention, laryngeal/pharyngeal edema), and upper extremities (edema). Commonly caused by malignancy (eg, mediastinal mass, Pancoast tumor) and thrombosis from indwelling catheters. Medical emergency. Can raise intracranial pressure (if obstruction is severe) → headaches, dizziness, ↑ risk of aneurysm/rupture of intracranial arteries.

**► RESPIRATORY—PHARMACOLOGY****Asthma drugs**

Bronchoconstriction is mediated by (1) inflammatory processes and (2) parasympathetic tone; therapy is directed at these 2 pathways.

Inhaled β_2 -agonists

Albuterol (short-acting), salmeterol, formoterol—relax bronchial smooth muscle. Can cause tremor, arrhythmia. Albuterol is for acute symptoms.

Inhaled or oral glucocorticoids

Fluticasone, budesonide—inhibit the synthesis of virtually all cytokines. Inactivate NF- κ B, the transcription factor that induces production of TNF- α and other inflammatory agents. 1st-line therapy for chronic asthma. Use a spacer or rinse mouth after use to prevent oral thrush.

Muscarinic antagonists

Tiotropium, ipratropium—competitively block muscarinic receptors, preventing bronchoconstriction. Also used for COPD. Tiotropium is long acting.

Antileukotrienes

Montelukast, zafirlukast—block leukotriene receptors (CysLT1). Especially good for aspirin-induced and exercise-induced asthma.

Zileuton—5-lipoxygenase inhibitor. ↓ conversion of arachidonic acid to leukotrienes. Hepatotoxic.

Anti-IgE monoclonal therapy

Omalizumab—binds mostly unbound serum IgE and blocks binding to Fc ϵ RI. Used in allergic asthma with ↑ IgE levels resistant to inhaled glucocorticoids and long-acting β_2 -agonists.

Methylxanthines

Theophylline—likely causes bronchodilation by inhibiting phosphodiesterase → ↑ cAMP levels due to ↓ cAMP hydrolysis. Limited use due to narrow therapeutic index (cardiotoxicity, neurotoxicity); metabolized by cytochrome P-450. Blocks actions of adenosine.

PDE-4 Inhibitors

Roflumilast—inhibits phosphodiesterase → ↑ cAMP → bronchodilation, ↓ airway inflammation. Used in COPD to reduce exacerbations.

Chromones

Cromolyn—prevents mast cell degranulation. Prevents acute asthma symptoms. Rarely used.